ELECTROTHON

PROBLEM STATEMENT

CHALLENGE-1

Part A (20 Points): 4-Bit ALU Core

Design a 4-bit ALU in Deeds with inputs A [3:0], B [3:0], and a 2-bit operation select OP[1:0]:

- 00: A + B (with Cout)
- 10: A AND B
- 10: A OR B
- 11: A logically shifted right by 1

Outputs: Result [3:0], Cout and a Overflow flag that goes high when result of addition exceeds 4 bits.
Try to use the minimum number of gates.
Bonus 10 Points if you do not use ready made blocks for Adders, MUXs, etc. and implement the entire ALU Core using Basic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) only.

Part B (30 Points): 4-Bit Synchronous Up-Down Counter

Design a 4-Bit Up-Down (Bidirectional) Binary Counter in Deeds with Reset and Mode (M) inputs. The counter must act as an Up counter (0 to 15) if M=0 and as a Down counter (15 to 0) if M=1. The reset must be synchronous.
You are allowed to use any of the flip-flops available in Deeds software.

CHALLENGE-2

Part A (30 Points): FSM 101

Design a circuit using Verilog HDL, that detects the sequence '101' in the last 4 inputs. Example: -

Input: 0 1 0 1 0 0 1 1 0 1 0 1 0

Output: 0 0 0 1 1 0 0 0 0 1 1 1 1

If you are unable to design the circuit so that it detects the sequence in the last 4 inputs, you can, for Partial Points (50% Reduction), design the circuit for an overlapping 101 sequence detector without the last 4 inputs condition.

Bonus 10 Points if you design circuits using both Moore and Mealy types of FSM.

Part B (50 Points): Frame Packet Receiver

You are to receive data packets over a 1-bit serial line in the following format:

| Start Bit (0) | 8 Data Bits | Even Parity Bit | Stop Bit (1) |

After receiving the stop bit, the line may stay at 1 in idle state before the next packet is transmitted.

Inputs:

- 1 Bit Input Stream
- Clock Signal
- Synchronous Reset

Outputs:

- done: high one-cycle pulse on successful reception.
- parity_err: high for one cycle if parity mismatches the received byte
- frame_err: high for one cycle if the stop bit is not 1.
- data_out[7:0]: most recently received byte, should be asserted with done signal and only if there are no frame or parity errors as described above, 0 otherwise.

Design a circuit in Verilog that takes a 1-bit stream as input and generates outputs as mentioned above..

Bonus Challenge: Command Receiver

Building upon the previous task, the packets are now in the format:

| Start Bit (0) | 2 Command Bits | 8 Data Bits | Even Parity Bit | Stop Bit (1) |

The circuit should have 2 registers A and B, 8 bits in length. The command bits describe what operation must be performed on the registers A and B. The output "result" depends on what the command is. Also note that the command must not be executed if there are any parity or frame errors. The commands are as follows:

- cmd = 00 : LOAD_A (Set A=Received Data Bits) (result = new value of A)
- cmd = 01 : LOAD_B (Set B=Received Data Bits) (result = new value of B)
- cmd = 10 : ADD (A+B) (result = A + B)
- cmd = 11 : CLEAR (Set A=B=0) (result = 0)

Inputs:

- 1 Bit Input Stream
- Clock Signal
- Synchronous Reset

Outputs:

- done: high one-cycle pulse on successful reception.
- parity_err: high for one cycle if parity mismatches the received byte.
- frame_err: high for one cycle if the stop bit is not 1.
- result [7:0]: display value as described by commands, should be asserted with done signal and only if there are no frame or parity errors as described above, 0 otherwise.

Design the above-described circuit in Verilog.

DELIVERABLES

- Deeds Files for Challenge 1 (Theory File if unable to make circuit for partial points (25% Reduction)).
- Verilog (.v) Files for Challenge 2 and Bonus Challenge
- PDF of FSM State Diagrams with Specifications of What Each State Means for Challenge 2 FSM's.

NOTE: Name the project in the following format:- DIGITAL_YOUR-ROLL NUMBER_NAME

RESOURCES

- https://www.youtube.com/watch?si=biyBoqMiH6BLdwB3EE101&v=P4TYU3FycQ I&feature=youtu.be (Deeds Tutorial).
- https://www.youtube.com/watch?v=paFaxtTCkI&list=PLiN_aCe7rPDepGl4TfOrzJNAalaw_qwE (Registers)
- https://www.youtube.com/watch?v=AaN72s5WfOM&list=PLuYnChSh1XdvuSGjQRi2jgUH9_CiVR8J&index=1 (Sequential Circuits)
- Counter (Neso Academy) - YouTube (Counters)
- https://www.youtube.com/playlist?list=PLiIOnHwN7NXw01eBDR7wI8KzGK4mu8Sr (Verilog Playlist)
- HDLBits (Verilog Practice Questions)
- Morris Mano Chapters 4-6
- You may also go through your 2nd semester Digital Theory Course Material.

NOTE: For the time being, you may use online simulators like EDA Playground to test out your Verilog codes but it is preferred to download and use Vivado for simulations.